Optimizing Deep Learning Models for Plant Leaf Disease Classification Using Nature-Inspired Algorithms

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Abstract. Plant diseases significantly affect agricultural productivity, particularly in island regions such as Mauritius, where timely diagnosis may be limited by access to expert knowledge. Image analysis based on deep learning became more popular in recently and it has emerged as a promising alternative to traditional manual inspection. This study proposes a plant leaf disease classification framework that combines a convolutional neural network with nature-inspired optimisation algorithms to enhance classification performance.
EfficientNet-B0 pretrained on ImageNet is adopted as the backbone model, and transfer learning is employed to adapt it to the target dataset. Multiple nature-inspired optimisation techniques are investigated to fine-tune key hyperparameters with the objective of improving generalisation and training efficiency. Experiments are conducted using the PlantVillage dataset, focusing on crop categories relevant to the Mauritian context, including tomato, potato, maize, and pepper. Standard preprocessing and data augmentation strategies are applied to strengthen model robustness.
Experimental results demonstrate that integrating nature-inspired optimisation methods can improve predictive performance, suggesting their potential value in supporting automated and scalable plant disease detection systems.

Keywords: Plant disease classification, deep learning, convolutional neural networks, EfficientNet, nature-inspired optimization, transfer learning, image-based diagnosis

1 Introduction

Agriculture is still an important field for food security and economic sustainability, particularly in developing and island nations such as Mauritius. However, crop production is frequently affected by plant diseases, which can significantly reduce yield and quality if not identified at an early stage. In many agricultural settings, disease diagnosis relies on manual examination of the plant by farmers or experts in agriculture. This process often takes a long time., subjective, and not always accessible to

smallholder farmers, especially in remote or resource-limited environments (Food and Agriculture Organization 2017; Savary et al. 2012).

With recent advances in artificial intelligence, computer vision has emerged as a promising tool for supporting agricultural decision-making. Image-based disease detection systems allow plant health to be assessed directly from leaf images, minimising the need for manual inspection. Among these approaches, deep learning models, particularly Convolutional Neural Networks (CNNs), have shown strong performance in visual recognition tasks by automatically learning relevant features from raw image data. Previous studies have shown that CNN-based models can effectively identify disease symptoms such as discoloration, lesions, and texture irregularities in plant leaves, making them suitable for automated plant disease classification (Sladojevic et al. 2016; Mohanty et al. 2016).

Even with these improvements, there are still some challenges. Many deep learning models require large amounts of labelled data and are prone to overfitting when trained on da-tasets collected under controlled conditions. To address these limitations, transfer learning and optimisation techniques have been increasingly adopted to improve model generalisation and training efficiency. In addition, the choice of preprocessing steps, regularisation strategies, and evaluation metrics plays an important role in de-termining overall system performance (Pan and Yang 2010; Too et al. 2019).

This research focuses on the categorisation of plant leaf diseases using deep learning techniques, with an emphasis on crops relevant to the Mauritian agricultural context, including tomato, potato, maize (corn), and pepper. The PlantVillage dataset is used as the primary data source due to its widespread use in plant disease research and the availability of well-labelled images. The task is formulated as a multi-class classification problem, where each input image is assigned a corresponding disease or healthy label (Hughes and Salathé 2015).

The proposed approach integrates a convolutional neural network–based framework with optimisation techniques inspired by natural processes to enhance model performance. In addition, systematic preprocessing and evaluation strategies are employed to assess classification accuracy and robustness. The suggested method's performance is analysed using standard evaluation metrics and comparative experiments.

This work has the following primary contributions:

a. the creation of a deep learning–based framework for plant disease classification tailored to crops relevant to Mauritius;
b. the incorporation of optimisation strategies to improve model performance; and
c. a comprehensive evaluation and comparative analysis of the proposed approach using benchmark data.

The rest of this paper is organised as follows. Section 2 shows the methods used in this study, including the convolutional neural network architecture, opti-misation strategy, dataset preparation, and evaluation metrics. The experimental results are shown and presented in Section 3. The last one, Section 4, concludes the paper and outlines directions for future work.

2 Methods

This research introduces a deep learning framework for the categorisation of plant diseases, combining a convolutional neural network with nature-inspired optimisation algorithms. The goal is to find out which nature-inspired algorithm works best for improving deep learning models that are used to identify plant diseases in images. These algorithms are specifically utilised to enhance hyperparameter tuning and model generalisation (Goodfellow et al. 2016).

The methodology is divided into four main parts. A convolutional neural network is the main model for sorting images. Second, a set of algorithms based on nature is used to find the best values for the most important hyperparameters of the deep learning model. Third, the steps for getting the dataset ready and doing some pre-processing are explained to make sure they are strong and consistent during training. Finally, we use standard performance evaluation metrics to test and compare the suggested optimisation strategies.

2.1 Convolutional Neural Network

Convolutional Neural Networks (CNNs) are used frequently for image classification jobs given their capacity to independently learn spatial and structural features from raw image data (Krizhevsky et al. 2012). Traditional machine learning methods depend on features that are designed by manually, but CNNs do both feature extraction and classification from start to finish. This makes them well suited for plant disease identification, where visual symptoms such as spots, lesions, and color variations appear at different spatial scales on leaf images.

A CNN usually consists of convolutional layers, activation functions, pooling layers (Goodfellow et al. 2016), and fully connected layers. In a convolutional layer, learnable filters are applied to the input image to retrieve local patterns. The convolution operation can be expressed as:

$$z^{\{(k)\}} = w^{\{(k)\}} * x + b^{\{(k)\}} \quad (1)$$

where x denotes the input image, $w^{(k)}$ represents the k-th convolutional filter, $b^{(k)}$ is the bias term, and $z^{(k)}$ denotes the resulting feature map produced by the convolution operation. The symbol * represents the convolution operator.

An activation function is used after convolution to make the network non-linear. This study uses the Rectified Linear Unit (ReLU) because it is simple and works well in deep neural networks (Nair and Hinton 2010). It is defined as:

$$f(z) = \max(0,z) \quad (2)$$

As the network depth increases, convolutional and pooling layers progressively extract higher-level feature representations. After that, these features are compressed and sent to fully connected layers for classification. The final output layer applies the

softmax function to transform the network outputs into class probabilities, ensuring that the predicted probabilities across all disease classes sum to one (Goodfellow et al. 2016). This enables multi-class plant disease classification by selecting the class with the highest predicted probability.

EfficientNet is a group of convolutional neural network architectures that use a compound scaling strategy to scale the depth, width, and input resolution of the network in a balanced way (Tan and Le 2019). This design enables EfficientNet models to achieve strong classification performance while maintaining computational efficiency. Among the different variants, EfficientNet-B0 represents the baseline architecture and provides an effective trade-off between accuracy and model complexity, making it suitable for image-based plant disease classification tasks.

In this study, EfficientNet-B0 pretrained on the ImageNet dataset is employed as the CNN backbone (Deng et al. 2009) as it has a good balance between classification accuracy and computational efficiency. Transfer learning is used by first freezing the pretrained layers and only training the classifier head to make the model fit the target dataset. Subsequently, all layers are unfrozen and fine-tuned using a smaller learning rate to further improve performance and generalisation on plant disease images.

Pooling Operation

Pooling layers are often used in CNN architectures to keep important information while lowering the spatial resolution of feature maps. This operation makes the computer work better and makes it less likely that translations will change the results. Pooling operations are built into the network blocks in EfficientNet, which helps with effective feature abstraction.

Training Objective

During training, the CNN's parameters are learned by minimising a classification loss function through gradient-based optimisation. This study employs cross-entropy loss alongside backpropagation to update the network weights, facilitating the model's acquisition of discriminative features for plant disease classification.

2.2 Nature-Inspired Optimisation Algorithms

To evaluate the impact of metaheuristic optimisation on CNN performance, this study investigates a diverse set of nature-inspired optimisation algorithms representing evolutionary and swarm intelligence approaches. These algorithms differ in their exploration–exploitation mechanisms and search dynamics, allowing a comprehensive comparison of their effectiveness in tuning hyperparameters for plant disease classification.

The chosen algorithms are Differential Evolution (DE), Particle Swarm Optimisation (PSO), Artificial Bee Colony (ABC), Grey Wolf Optimiser (GWO), Bat Algorithm (BA), Firefly Algorithm (FA), Cuckoo Search (CS), and Coral Reef Optimisation (CRO). People often use DE and PSO to solve problems that need to be optimised all

the time, like tuning the learning rate and weight decay. GA has a strong ability to search the whole world thanks to crossover and mutation operations. GWO strikes a good balance between exploration and exploitation. BA, FA, and CS bring different search dynamics that help keep things from converging too quickly and finding local optima. CRO, on the other hand, increases population diversity through competitive reproduction mechanisms.

These algorithms are implemented using the NiaPy framework and applied to optimise selected hyperparameters of the EfficientNet-B0 model. Their comparative performance is analysed in Section 3.

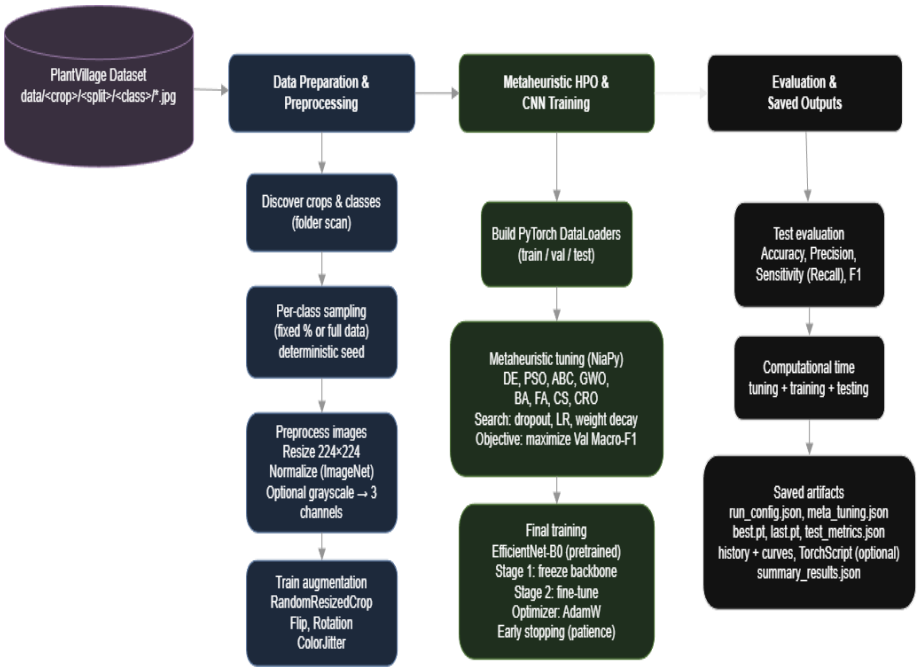

Fig. 1. Architecture of the Plant Disease Detection using NIA

2.3 Dataset

The Plant Village Dataset (Updated) is a large image collection designed for plant disease detection and classification. It contains around 70,000 high-quality photos of plant leaves, covering both healthy and diseased examples across nine different plant species. The dataset is organised in a very practical way for machine learning: each crop is stored in its own folder and then split into Train, Validation, and Test subsets, with the same set of category labels appearing across these splits. Within each split, images are grouped into class folders (for example, “Healthy” versus specific disease

names), which makes it straightforward to load the data using standard folder-based pipelines in PyTorch.

Table 1. Distribution of plant disease images across different crop categories.

Crop	Train	Validation	Test	Total	Split ratio (Train/Val/Test)
Apple	7,771	1,747	196	9,714	80.00%/ 17.98% / 2.02%
Tomato	11,105	2,495	280	13,880	80.01%/ 17.98% / 2.02%
Potato	5,702	1,282	144	7,128	79.99%/ 17.99% / 2.02%
Corn	7,316	1,645	184	9,145	80.00%/ 17.99% / 2.01%
Pepper	3,901	877	98	4,876	80.00%/ 17.99% / 2.01%
Grand total	**35,795**	**8,046**	**902**	**44,743**	**80.00%/ 17.98% / 2.02%**

Figure 2 - Samples images per class shows sample images from the Tomato Training Dataset across all six classes, illustrating the visual differences the model learns to distinguish.

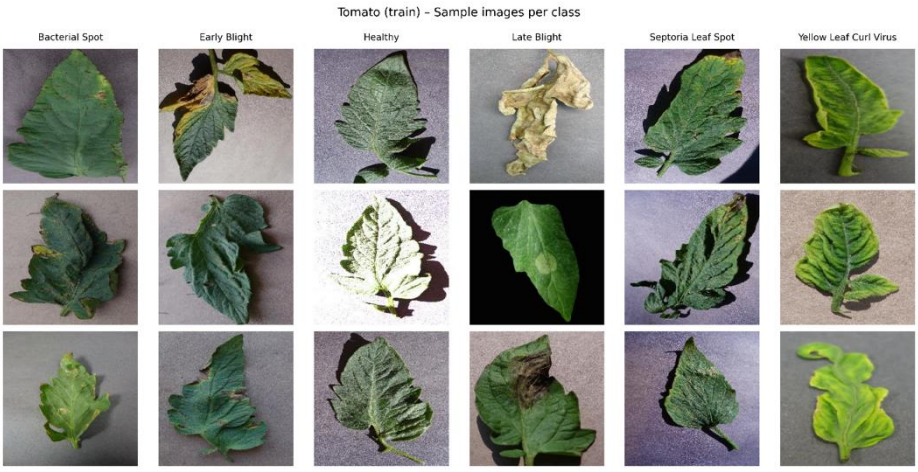

Fig. 2. illustrates representative training samples from the tomato dataset, showing visual differences across disease classes.

3 Data Preprocessing

In preprocessing, every leaf image is loaded and standardised so the model always sees the same kind of input. The pipeline is kept simple and consistent for testing and validation. Images are resized to 224 × 224 pixels, turned into tensors, and then normalised using the ImageNet mean and standard deviation. This is important because the EfficientNet-B0 backbone was trained on ImageNet, so using the same input scaling helps the pretrained features work as they should and keeps training stable (Deng et al. 2009; Tan and Le 2019).

For the training set, we use the same core steps but add light, realistic augmentation to improve generalisation. Instead of a fixed resize, we apply a random resized crop so the model learns to handle small changes in framing while still keeping most of the leaf visible. We also include horizontal flips and small rotations (up to 15°) to reflect how leaves naturally appear in different orientations. On top of that, we apply gentle colour jitter to mimic natural variation in lighting and camera conditions. For the grayscale experiment, images are converted to grayscale while keeping three channels, which allows us to use the pretrained EfficientNet without modifying its input layer. In this case, the jitter is limited to brightness and contrast to stay consistent with grayscale inputs. Overall, these preprocessing and augmentation choices increase the variety of training examples without changing the underlying class, helping reduce overfitting and improving performance on unseen test images (Krizhevsky et al. 2012; Shorten and Khoshgoftaar 2019).

4 Performance Metrics

The classification metrics used in this study are described below.

Accuracy: The performance metric that measures the proportion of correct predictions the model makes across the entire test set.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (3)$$

Precision: The proportion of cases the model labels as positive that are actually positive. It shows how well the model reduces false positives (that is, avoids false alarms).

$$Precision = \frac{TP}{(TP + FP)} \quad (4)$$

Recall / Sensitivity (True Positive Rate) : The percentage of real positives that have been identified correctly. It shows the extent to which the model finds all the right cases and doesn't miss any (false negatives).

$$Recall = \frac{TP}{(TP + FN)} \quad (5)$$

F1-score: The average of precision and recall. It is especially helpful when class frequencies aren't perfectly balanced because it rewards models that find a good balance between precision and recall instead of just optimising one of them.

$$F1 = 2 * \frac{(Precision * Recall)}{(Precision + Recall)} \quad (6)$$

Where **TP** is *true positives*, **TN** is *true negatives*, **FN** is *false negatives*, and **FP** is *false positives*.

5 Results & Discussion

Table 2 presents a summary of the results for all metaheuristic–CNN experiments performed in this study, where the same EfficientNet-B0 (pretrained) backbone was used throughout and the metaheuristics were applied to tune key training hyperparameters (dropout, learning rates, and weight decay) using validation Macro-F1 as the objective. Overall, all approaches achieved very strong performance (above 99% across the main metrics), which shows that transfer learning with EfficientNet-B0 is highly effective for this PlantVillage classification setup.

Firstly, Differential Evolution (DE) produced a strong baseline with 99.21% accuracy and a 99.21% Macro-F1 score, completing in approximately 32.97 minutes. Building on this, Particle Swarm Optimisation (PSO) delivered a noticeable improvement, reaching 99.37% accuracy and 99.36% Macro-F1, while also being one of the most time-efficient runs at around 29.15 minutes. Furthermore, the Grey Wolf Optimiser (GWO) achieved the best overall results in Table 2, recording the highest accuracy (99.45%) and Macro-F1 (99.46%), which indicates the most consistent performance across classes, with a total runtime of roughly 31.19 minutes.

Moreover, both the Bat Algorithm (BAT) and Cuckoo Search (CS) achieved very similar mid-range performance (about 99.29% accuracy and 99.28% Macro-F1), although BAT was slightly faster overall (29.93 minutes) compared with CS (32.29 minutes). The Artificial Bee Colony (ABC) run delivered results comparable to the DE baseline (around 99.21% accuracy and 99.20% Macro-F1) with a total time of 30.35 minutes. In contrast, the Firefly Algorithm (FA) produced the lowest Macro-F1 in Table 2 (99.18%) and required a longer total runtime (34.15 minutes), suggesting that under the same tuning budget it was less effective at locating the best hyperparameter region.

Finally, the Coral Reefs Optimisation (CRO) approach achieved performance that was very close to PSO (99.36% Macro-F1) but at a substantially higher computational cost (45.66 minutes), making it a less practical choice when runtime is a concern. In

summary, GWO provided the best overall predictive performance in this study, while PSO offered a very competitive alternative with excellent results and the most efficient runtime among the top-performing methods.

Table 2. Performance comparison of EfficientNet-B0 combined with different nature-inspired optimisation algorithms.

Model	Computational (min)	Accuracy	Precision (Positive predicted value)	Recall / Sensitivity (True positive rate)	F1-Score
EfficientNet-B0 + DE	32.97	0.9921	0.9924	0.9920	0.9921
EfficientNet-B0 + PSO	29.15	0.9937	0.9940	0.9934	0.9936
EfficientNet-B0 + GWO	31.19	0.9945	0.9949	0.9944	0.9946
EfficientNet-B0 + BAT	29.93	0.9929	0.9929	0.9927	0.9928
EfficientNet-B0 + Firefly (FA)	34.15	0.9921	0.9925	0.9915	0.9918
EfficientNet-B0+ Cuckoo Search (CS)	32.29	0.9929	0.9929	0.9927	0.9928
EfficientNet-B0 + CRO	45.66	0.9937	0.9938	0.9935	0.9936
EfficientNet-B0 + ABC	30.35	0.9921	0.9921	0.9921	0.9920

5.1 Comparative Analysis

Table 3. According to the results in Table 3, our suggested EfficientNet-BO model with metaheuristic tuning (GWO) has the best overall performance, with a test accuracy of 99.45%. This shows that the selected model works better than other methods for detecting plant diseases from images when it comes to classifying them.

Table 3. Comparative analysis of existing models

Reference	Algorithm	Accuracy (%)
Mohanty et al. (2016)	CNN	99.35
Sladojevic et al. (2016)	CNN	96.30
Picon et al. (2019)	CNN	94.80
Zhang et al. (2019)	CNN	94.65
Fuentes et al. (2017)	CNN	96.30
Chosen Hybrid Model	**EfficientNet-BO + GWO**	**99.45 %**

6 Conclusion & Future Works

This study shows that nature-inspired optimisation algorithms have strong potential to support plant disease detection and, in turn, improve crop health and productivity in

Mauritius. By enabling faster and more accessible diagnosis from images—without always relying on expert inspection—such systems can help farmers respond earlier, reduce unnecessary losses, and potentially improve profitability. In a wider context, this direction aligns well with SDG 2 (Zero Hunger) by contributing to more resilient and efficient agricultural practices.

Among all approaches evaluated, the most effective configuration combined EfficientNet-B0 with Grey Wolf Optimiser (GWO) to tune key hyperparameters, namely dropout, learning rates, and weight decay. The strong test performance achieved suggests that metaheuristic-driven tuning can meaningfully improve model generalisation. In addition, the preprocessing strategy—resizing images to a consistent input size and applying light, realistic augmentation such as horizontal flips and small rotations—helped the model become more robust to variations in viewpoint and leaf orientation, which is important for real-world use where conditions are rarely controlled.

For future work, further validation is needed to confirm robustness across a broader range of crops, disease types, and image conditions (e.g., lighting, background clutter, blur, and camera quality). Expanding the dataset to include more crops and more diverse field images would also help the model generalise beyond curated datasets. Finally, it would be valuable to explore deployment-focused improvements such as lightweight inference for mobile devices, explainability tools to build user trust, and continual learning strategies so the system can be updated as new disease patterns emerge.

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